Now we can take the marginal probability $\nu(\underline{\sigma}_{\setminus u})$ out of the expression within the summations in the LHS. Then we can use the definition of conditionals given by: $\nu(\sigma_i|\sigma_{\setminus i}) = \frac{\nu(\underline{\sigma})}{\nu(\underline{\sigma}_{\setminus i})}$, $\nu(\alpha|\sigma_{\setminus i}) = \frac{\nu(\underline{\sigma}_{u\to\alpha})}{\nu(\underline{\sigma}_{\setminus i})}$, to rewrite it as follows,

$$\frac{1}{2}\sum_{\underline{\sigma}\in\mathcal{S}}\sum_{u=1}^{n}\sum_{\alpha\in Q,\alpha\neq\sigma_i}\frac{P(\underline{\sigma}_{u\to\alpha}|\underline{\sigma})}{\mu(\alpha|\underline{\sigma}_{\setminus u})}\nu(\underline{\sigma}_{\setminus u})\Big|\mu(\alpha|\underline{\sigma}_{\setminus u})\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(\alpha|\underline{\sigma}_{\setminus u})\Big| \leq \eta. \tag{D5}$$

Now applying Condition 1 we can write this as,

$$\frac{1}{2}\sum_{\underline{\sigma}\in\mathcal{S}}\sum_{u=1}^{n}\nu(\underline{\sigma}_{\setminus u})\sum_{\alpha\in Q,\alpha\neq\sigma_i}\Big|\mu(\alpha|\underline{\sigma}_{\setminus u})\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(\alpha|\underline{\sigma}_{\setminus u})\Big| \leq \frac{\eta}{\omega_P}. \tag{D6}$$

Now we assert that the expression $\sum_{\alpha\in[Q],\alpha\neq\sigma_i}\Big|\mu(\alpha|\underline{\sigma}_{\setminus u})\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(\alpha|\underline{\sigma}_{\setminus u})\Big|$ is lower bounded simply by $|\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})|$.

To show this consider two vectors $a, b \in \mathbb{R}^{|Q|}$, such that they sum to one, $\sum_i a_i = \sum_i b_i = 1$. It is simple to see that,

$$|a_k - b_k| = |a_k\sum_{j\in Q} b_j - b_k\sum_{j\in Q} a_j| = |\sum_{j\in Q, j\neq k} a_k b_j - a_j b_k| \leq \sum_{j\in Q, j\neq k}|a_k b_j - a_j b_k| \tag{D7}$$

Since the conditionals sum to one, as a consequence of the above inequality we can see that, $\sum_{\alpha\in Q,\alpha\neq\sigma_i}\Big|\mu(\alpha|\underline{\sigma}_{\setminus u})\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\nu(\alpha|\underline{\sigma}_{\setminus u})\Big| \geq |\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})|$. Putting this into (D6), we get,

$$\frac{1}{2}\sum_{u=1}^{n}\sum_{\underline{\sigma}\in\mathcal{S}}\nu(\underline{\sigma}_{\setminus u})\Big|\nu(\sigma_u|\underline{\sigma}_{\setminus u}) - \mu(\sigma_u|\underline{\sigma}_{\setminus u})\Big| \leq \frac{\eta}{\omega_P}. \tag{D8}$$

Now by splitting the inner sum over $\underline{\sigma}$ as one going over σ_u and another over $\underline{\sigma}_{\setminus u}$, we can easily deduce that,

$$\sum_{u=1}^{n}\sum_{\underline{\sigma}\in\mathcal{S}}\nu(\underline{\sigma})\Big|\nu(.|\underline{\sigma}_{\setminus u}) - \mu(.|\underline{\sigma}_{\setminus u}))\Big|_{TV} \leq \frac{\eta}{\omega_P}. \tag{D9}$$

□

Appendix E: Proof of corollary 2

Proof. For a fixed $u \in [n]$ and parametric conditionals are given by $p(q|\underline{\sigma}_{\setminus u};\underline{\theta}) = \frac{1}{1+\sum_{p\in Q,p\neq q} e^{E(\underline{\sigma}_{u\to p},\underline{\theta}) - E(\underline{\sigma}_{u\to q},\underline{\theta})}}$. And the corresponding log-liklihoods are $\mathcal{L}_u(\underline{\theta},\underline{\sigma}) = -\log(p(\sigma_u|\underline{\sigma}_{\setminus u};\underline{\theta}))$ The average KL divergence between the conditionals of ν and the parametric hypothesis can be written as,

$$\mathop{\mathbb{E}}_{\underline{\sigma}_{\setminus u}\sim\nu_{\setminus u}} D_{KL}\left(\nu(.|\underline{\sigma}_{\setminus u})||p(.|\underline{\sigma}_{\setminus u};\underline{\theta}_u)\right) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu}\log\left(\nu(\sigma_u|\underline{\sigma}_{\setminus u})(1+\sum_{p\in Q,p\neq q} e^{E(\underline{\sigma}_{u\to p},\underline{\theta}) - E(\underline{\sigma}_{u\to q},\underline{\theta})})\right) \tag{E1}$$

$$= \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu}\log\Big(\nu(\sigma_u|\underline{\sigma}_{\setminus u})\Big) + \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu}\mathcal{L}_u(\underline{\theta},\underline{\sigma}) \tag{E2}$$

For convenience, let us define ρ as the lower bound on the conditionals of the parametric distribution, which is assumed to also include the true distribution, i.e. $\rho \equiv \min_{\sigma,\underline{\theta},u} p(\sigma_u|\underline{\sigma}_{\setminus u};\underline{\theta})$.

From the reverse Pinsker's inequality (Lemma 4.1 in [21]) and the lower bound on the conditionals we see that, $D_{KL}(\nu(.|\sigma_{\setminus u})||\mu(.|\sigma_{\setminus u})) \leq \frac{2}{\rho}\left|\nu(.|\underline{\sigma}_{\setminus u}) - \mu(.|\underline{\sigma}_{\setminus u})\right|_{TV}$. Using this bound in Theorem 1 gives us,

$$\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \log(\nu(\sigma_u|\sigma_{\setminus u})) + \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\underline{\theta}^*_u, \underline{\sigma}) \leq \frac{2\eta}{\rho\ \omega_P}. \tag{E3}$$

Now from the positivity of KL divergence, $\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \log(\nu(\sigma_u|\sigma_{\setminus u})) \geq \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \log(p(\sigma_u|\sigma_{\setminus u};\hat{\underline{\theta}}_u)) = -\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\hat{\underline{\theta}}_u, \underline{\sigma})$. This gives us,

$$\mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\underline{\theta}^*_u, \underline{\sigma}) - \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\hat{\underline{\theta}}_u, \underline{\sigma}) \leq \frac{2\eta}{\rho\ \omega_P} \tag{E4}$$

Now the bounds on the parametric conditionals can be used to bound the likelihood $\mathcal{L}(\theta, \underline{\sigma})$ as follows, $\log(\frac{1}{1-\rho}) \leq \mathcal{L}(\theta, \underline{\sigma}) \leq \log(\frac{1}{\rho})$. This establishes $\mathcal{L}(\theta, \sigma)$ as a bounded random variable. Now we can use Hoeffding's inequality (Lemma eflem:hoeff) to show that with probability $1-\delta$,

$$\frac{1}{M'}\left(\sum_{t=1}^{M'} \mathcal{L}(\underline{\theta}^*_u, \underline{\sigma}^{(t)}) - \sum_{t=1}^{M'} \mathcal{L}(\hat{\underline{\theta}}_u, \underline{\sigma}^{(t)})\right) \leq \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\underline{\theta}^*_u, \underline{\sigma}) - \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\hat{\underline{\theta}}_u, \underline{\sigma}) + \log\left(\frac{1-\rho}{\rho}\right)\sqrt{\frac{\log(1/\delta)}{2M'}}, \tag{E5}$$

$$\leq \frac{2\eta}{\rho\omega_P} + \log\left(\frac{1-\rho}{\rho}\right)\sqrt{\frac{\log(1/\delta)}{2M'}}. \tag{E6}$$

Now from (16) we see that $\rho \geq \frac{1}{1+(|Q|-1)e^{2\gamma}}$. Plugging this into the above equation gives us the desired result.

□

Appendix F: Proofs of Learning guarantees

For Ising models, the exact PL estimator and the one computed from samples have the following form

$$\mathcal{L}(\underline{\theta}_u) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \mathcal{L}(\underline{\theta}_u, \underline{\sigma}) = \mathop{\mathbb{E}}_{\underline{\sigma}\sim\nu} \log(1 + \exp(-2(\sum_{k\neq u} \theta_{u,k}\sigma_u\sigma_k + \theta_u))) \tag{F1}$$

$$\mathcal{L}_M(\underline{\theta}_u) = \frac{1}{M}\sum_{t=1}^{M} \mathcal{L}(\underline{\theta}_u, \underline{\sigma}^{(t)}) = \frac{1}{M}\sum_{t=l}^{M} \log(1 + \exp(-2\sigma_u^{(t)}(\sum_{k\neq u} \theta_{u,k}\sigma_k^{(t)} + \theta_u))) \tag{F2}$$

$$\hat{\underline{\theta}}_u \coloneqq \mathop{\mathrm{argmin}}_{||\underline{\theta}_u||_1 \leq \gamma} \mathcal{L}_M(\underline{\theta}_u). \tag{F3}$$

A key quantity we will use through out this proof is the following measure of curvature of the loss around $\underline{\theta}^*_u$

$$\delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) \coloneqq \mathcal{L}_M(\hat{\underline{\theta}}_u) - \mathcal{L}_M(\underline{\theta}^*_u) - \langle\Delta, \nabla\mathcal{L}_M(\underline{\theta}^*_u)\rangle \tag{F4}$$

a. Proof of Theorem 3

Proof. First let us define two error vectors that respectively capture the error in all parameters connected to a variable at u and only the error in the pairwise terms connected to u.

$$\Delta \coloneqq \hat{\underline{\theta}}_u - \underline{\theta}^*_u \in \mathbb{R}^n, \tag{F5}$$

$$\Delta_{\setminus u} \coloneqq \hat{\underline{\theta}}_{\setminus u} - \underline{\theta}^*_{\setminus u} \in \mathbb{R}^{n-1}. \tag{F6}$$

That is, these two error vectors only differ by $\hat{\theta}_u - \theta^*_u$.

Now from the definition of the curvature function in (F4),

$$\mathcal{L}_M(\hat{\underline{\theta}}_u) - \mathcal{L}_M(\underline{\theta}^*_u) = \delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) + \langle \Delta, \nabla\mathcal{L}_M(\underline{\theta}^*_u)\rangle, \tag{F7}$$

$$\geq \delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) - \left|\langle \Delta, \nabla\mathcal{L}_M(\underline{\theta}^*_u)\rangle\right|, \tag{F8}$$

$$\geq \delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) - ||\Delta||_1||\nabla\mathcal{L}_M(\underline{\theta}^*_u)||_\infty \tag{F9}$$

Now to avoid a contradiction with the definition of $\hat{\underline{\theta}}$ as the minimizer of $\mathcal{L}_M$, the quantity on the left must be negative. This then implies that $\delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) \leq ||\Delta||_1||\nabla\mathcal{L}_M(\underline{\theta}^*_u)||_\infty$. Due to the ℓ_1 constraint in the problem we have $||\Delta||_1 \leq 2\gamma$. This gives us,

$$\delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) \leq 2\gamma||\nabla\mathcal{L}_M(\underline{\theta}^*_u)||_\infty. \tag{F10}$$

Now we have two technical lemmas that give relevant bounds that can turn the above inequality in to a learning guarantee.

From Lemma 3, when $M \geq \frac{8}{\varepsilon_a^2}\log(\frac{4n^2}{\delta})$ the following statement holds with probability $1 - \frac{\delta}{2n}$,

$$||\nabla\mathcal{L}_M(\underline{\theta}^*_u)||_\infty \leq \frac{4\eta}{\omega_P} + \varepsilon_a \tag{F11}$$

From Lemma 6, when $M \geq \frac{2\gamma^4}{\varepsilon_b^2}\log(\frac{2n}{\delta})$, the following statement holds with probability $1 - \frac{\delta}{2n}$,

$$\delta\mathcal{L}_M(\Delta, \underline{\theta}^*_u) \geq \frac{e^{-4\gamma}}{2}||\Delta_{\setminus u}||^2_\infty - \frac{4e^{-2\gamma}\eta}{\omega_P} - \varepsilon_b \tag{F12}$$

Using these bounds in (F10) gives us the following bound with probability $1 - \delta/n$,

$$||\Delta_{\setminus u}||^2_\infty - \frac{8\eta e^{2\gamma}}{\omega_P} - 2e^{4\gamma}\varepsilon_b \;\leq\; 4e^{4\gamma}\gamma\left(\frac{4\eta}{\omega_P} + \varepsilon_a\right). \tag{F13}$$

Now make the following choices; choose ε_a and ε_b such that $4e^{4\gamma}\gamma\varepsilon_a = 2e^{4\gamma}\varepsilon_b = \varepsilon^2/2$, and choose $M = \left\lceil 2^9 \frac{e^{8\gamma}\gamma^4}{\varepsilon^4}\log(\frac{8n^2}{\delta})\right\rceil \geq \max\left(\frac{2\gamma^4}{\varepsilon_b^2}\log(\frac{2n}{\delta}), \frac{8}{\varepsilon_a^2}\log(\frac{8n^2}{\delta})\right)$. Plugging these choices in the expression above gives us the following bound with probability $1 - \delta/n$.

$$||\Delta_{\setminus u}||^2_\infty \leq \frac{8\eta e^{2\gamma}}{\omega_P} + \frac{16e^{4\gamma}\gamma\eta}{\omega_P} + \varepsilon^2, \tag{F14}$$

$$\leq \frac{16\eta(1+\gamma)e^{4\gamma}}{\omega_P} + \varepsilon^2. \tag{F15}$$

$$\tag{F16}$$